

Autonomous Forest Data Collection
Simulating a Multi-Agent C-ITS System

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■ Motivation

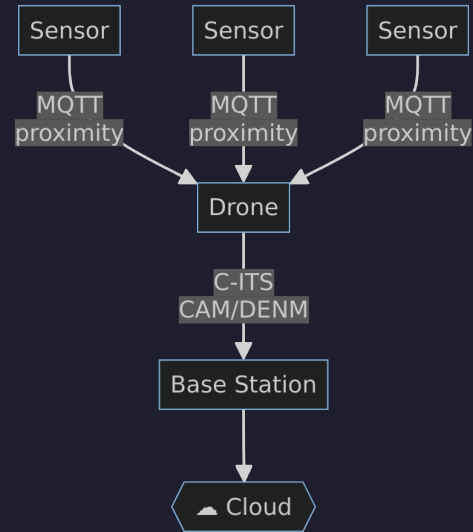
Forest monitoring requires **continuous, large-scale coverage** of remote terrain.

Human inspection is costly, slow, and dangerous.

Idea: Deploy autonomous drones that sweep a forest area, detect ground sensors, collect environmental readings, and relay data to a base station – all coordinated via **standardised vehicular communication (ETSI C-ITS)**.

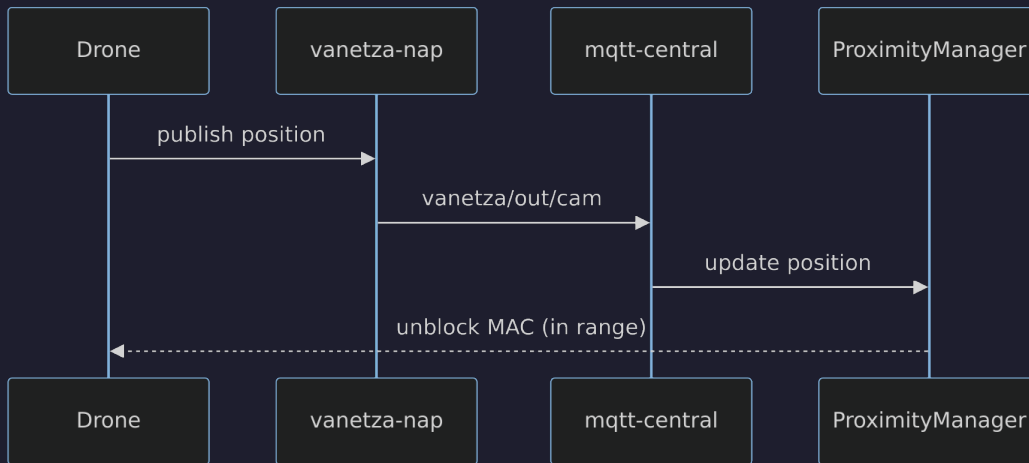
System Overview

- Sensors are **static** – announce position, serve data on request via MQTT
- Drones **sweep** the area, detect sensors via proximity, collect readings
- Drone ↔ Base station communicate via **C-ITS** (CAM/DENM)
- Base station **aggregates** data and uploads to cloud



■ CAM – Cooperative Awareness Messages

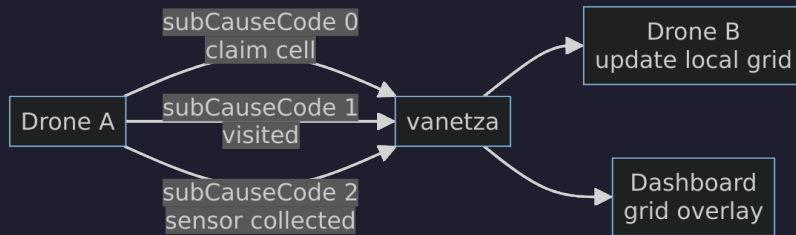
Purpose: every entity continuously broadcasts its presence and position.



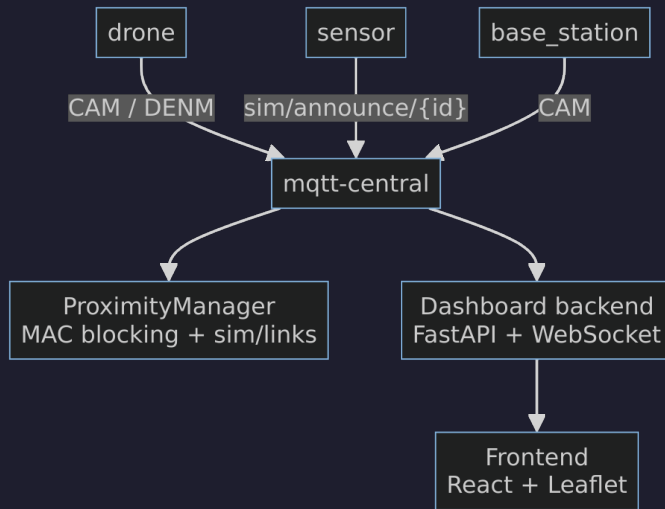
- Drone detects a **sensor** in range → CAM with `stationType = 20`
- Drone detects **base station** → CAM with `stationType = 15`

■ DENM – Decentralised Environmental Notification Messages

Purpose: drones publish cell events as they sweep the area.



- Peers avoid re-visiting already **claimed/visited** cells
- Cell index carried in `actionId.sequenceNumber`
- On first contact between two drones → full **grid sync** via `sim/grid_sync/{id}` (MQTT)



| <http://localhost:3000>

- Map with base station, sensors, and connectivity links
- Entity panel – station IDs, coordinates, tick counter
- Links appear/disappear as entities move in/out of range

make up

■ What Is Implemented

- **Network baseline** – vanetza-nap containers exchanging CAMs over a virtual ITS-G5 network
- **ProximityManager** – dynamic MAC blocking/unblocking based on simulated range
- **Sensors** – self-assign identity at startup, announce via `sim/announce/{id}` (retained), serve data on request
- **Drone core** – identity bootstrap, CAM publishing, state machine stub
- **Base station** – announces itself, receives data deliveries
- **Dashboard** – real-time map, entity list, connectivity links via WebSocket

■ What Is Missing

- **Drone movement** – position is currently static; boustrophedon path not yet implemented
- **Coverage grid** – cell state tracking (UNKNOWN / CLAIMED / VISITED / SENSOR_FOUND)
- **Sensor collection flow** – drone detecting sensor in range and requesting data
- **Multi-drone coordination** – DENM cell events + MQTT grid sync between peers
- **Mission completion** – base station aggregating all data and uploading to cloud
- **Dashboard grid overlay** – rendering cell states on the map

■ Thank You

```
make prepare-env    # copy .env-sample → .env  
make up             # start simulation  
# open http://localhost:3000
```

Questions?